EC301 — Electronic Devices

Semester III • 3L:0T:0P • 3 Credits

Module 3 Exam Notes: MOSFET Physics, C-V & Small Signal Models

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. The MOS Capacitor Physics and Energy Bands

The **Metal-Oxide-Semiconductor (MOS) capacitor** is the fundamental two-terminal building block of all MOS Field-Effect Transistors. It consists of a top metal (or heavily doped polycrystalline silicon) gate electrode, an insulating silicon dioxide (SiO_2) layer of thickness t_{ox}, and a semiconductor silicon substrate.

1.1 Basic Structure and Flat-Band Condition

Consider an ideal MOS capacitor on a p-type Silicon substrate with acceptor doping N_A:

- **Oxide Capacitance per unit area (C_{ox}):**

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} \tag{1}$$

  where $\epsilon_{ox} = \epsilon_r \epsilon_0 \approx 3.9 \times 8.854 \times 10^{-14}\,\text{F/cm} \approx 3.45 \times 10^{-13}\,\text{F/cm}$.

- **Flat-Band Voltage (V_{FB}):** Due to the difference in work function between the gate metal Φ_m and semiconductor Φ_s ($\Phi_{ms} = \Phi_m - \Phi_s$), and fixed positive charges trapped inside the oxide or at the oxide-semiconductor interface (Q_{ox}), the energy bands are bent even at zero gate bias ($V_G = 0$). The specific external gate voltage that restores flat energy bands up to the interface ($\psi_s = 0$) is the **flat-band voltage**:

$$V_{FB} = \Phi_{ms} - \frac{Q_{ox}}{C_{ox}} \tag{2}$$

1.2 Three Surface Charge Regimes of the MOS Capacitor

As the gate voltage V_G varies relative to V_{FB}, the surface potential ψ_s at the Si–SiO_2 interface changes, establishing three distinct operating regimes:

1. **Accumulation ($V_G < V_{FB}$ for p-substrate):** A negative gate voltage bends the energy bands upward at the interface ($\psi_s < 0$), pulling the valence band edge E_v closer to the Fermi level E_F. Majority holes accumulate at the silicon surface, forming a thin sheet of positive mobile hole charge ($Q_s > 0$).

2. **Depletion ($V_{FB} < V_G < V_T$ for p-substrate):** A positive gate voltage bends the energy bands downward ($\psi_s > 0$), pushing majority holes away from the surface into the bulk. This leaves behind an uncompensated, immobile space charge layer of ionized acceptor atoms ($-qN_A$) of depletion width W:

$$W = \sqrt{\frac{2\epsilon_s \psi_s}{qN_A}} \tag{3}$$

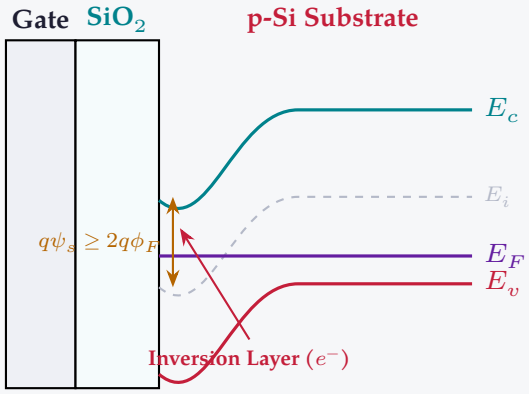
   The depletion charge per unit area is $Q_d = -qN_A W = -\sqrt{2q\epsilon_s N_A \psi_s}$.

3. **Inversion ($V_G \geq V_T$ for p-substrate):** As V_G increases further, the bands bend downward so severely that the intrinsic Fermi level E_i crosses below the Fermi level E_F at the interface.

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- **Onset of Weak Inversion:** Occurs when E_i crosses E_F at the surface ($\psi_s = \phi_F$), where $\phi_F = V_T \ln(N_A/n_i)$ is the bulk potential. The surface becomes n-type ($n_s > p_s$).
- **Threshold / Strong Inversion ($\psi_s = 2\phi_F$):** The electron concentration at the surface equals the original bulk hole concentration: $n_s = N_A$. An extremely thin (< 5–$10\,\text{nm}$) and highly conductive layer of minority electrons (the **inversion layer**) forms directly beneath the oxide.

Energy Band Diagram of MOS Capacitor on P-Substrate in Inversion



1.3 Threshold Voltage Derivation

At the onset of strong inversion ($\psi_s = 2\phi_F$), the depletion region reaches its maximum width W_{max}:

$$W_{\text{max}} = \sqrt{\frac{4\epsilon_s \phi_F}{qN_A}} \tag{4}$$

Beyond this point, any further increase in gate voltage is shielded by the exponentially growing inversion electron charge, locking the depletion width at W_{max}. The maximum depletion charge per unit area is:

$$Q_{d,\text{max}} = -qN_A W_{\text{max}} = -\sqrt{4q\epsilon_s N_A \phi_F} \tag{5}$$

By voltage loop across the MOS structure, the applied gate threshold voltage V_T is:

$$V_T = V_{FB} + V_{ox} + \psi_s = V_{FB} - \frac{Q_{d,\text{max}}}{C_{ox}} + 2\phi_F \tag{6}$$

Threshold Voltage and Body Effect Equations

$$V_{T0} = V_{FB} + 2\phi_F + \frac{\sqrt{4q\epsilon_s N_A \phi_F}}{C_{ox}} \tag{7}$$

When a reverse substrate-to-source body bias ($V_{SB} > 0$) is applied:

$$V_T = V_{T0} + \gamma\left(\sqrt{2\phi_F + V_{SB}} - \sqrt{2\phi_F}\right) \tag{8}$$

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where γ is the **body effect coefficient**:

$$\gamma = \frac{\sqrt{2q\epsilon_s N_A}}{C_{ox}} \quad (\mathrm{V}^{1/2}) \tag{9}$$

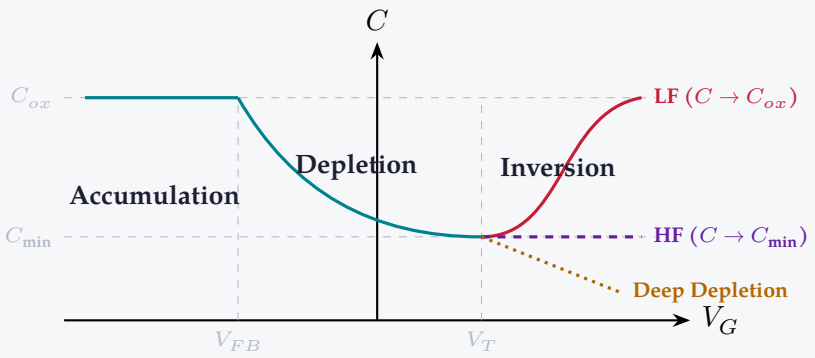
2. C-V Characteristics of the MOS Capacitor

The capacitance of an MOS capacitor is defined as $C = dQ_G/dV_G$. The total capacitance is the series combination of the fixed oxide capacitance C_{ox} and the semiconductor surface capacitance C_s:

$$C = \frac{C_{ox}C_s}{C_{ox} + C_s} \tag{10}$$

Bias Regime	Low-Frequency (LF) Capacitance	High-Frequency (HF) Capacitance
Accumulation ($V_G < V_{FB}$)	$C = C_{ox}$ Majority holes accumulate immediately at interface ($C_s \to \infty$).	$C = C_{ox}$ Majority holes respond rapidly up to multi-GHz frequencies.
Depletion ($V_{FB} < V_G < V_T$)	$C = \frac{C_{ox}C_{\text{dep}}}{C_{ox}+C_{\text{dep}}} < C_{ox}$ Decreases monotonically as depletion layer widens ($W \propto \sqrt{\psi_s}$).	$C = \frac{C_{ox}C_{\text{dep}}}{C_{ox}+C_{\text{dep}}} < C_{ox}$ Depletion layer capacitance dominates identically for LF and HF.
Strong Inversion ($V_G \geq V_T$)	$C = C_{ox}$ Thermal generation-recombination (1–100 Hz) keeps pace with slow AC signal; inversion charge Q_n modulates.	$C = C_{\text{min}} = \frac{C_{ox}C_{\text{dep,max}}}{C_{ox}+C_{\text{dep,max}}}$ Minority carriers cannot respond ($f \geq 100$ kHz); depletion layer stays locked at W_{max}.
Deep Depletion (Fast Ramp)	Not observed under slow quasi-static sweeps.	C continues to decrease below C_{min} because inversion layer has no time to generate; $W > W_{\text{max}}$.

MOS Capacitor Low-Frequency and High-Frequency C-V Curves



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3. MOSFET Operation and I-V Characteristics

The MOSFET is a four-terminal, voltage-controlled majority carrier field-effect transistor. By utilizing heavily doped Source and Drain regions that act as reservoirs of mobile electrons, the inversion channel can be modulated instantly, bypassing the slow thermal generation limit of the MOS capacitor.

3.1 Operating Regimes of an n-Channel Enhancement MOSFET

1. **Cut-off Region ($V_{GS} < V_T$):** No continuous inversion channel connects the source and drain. The drain-to-source path comprises two back-to-back p-n junctions (n^+–p–n^+). Ideal current $I_D = 0$. (In practice, subthreshold diffusion current flows: $I_D \propto \exp\left(\frac{V_{GS}-V_T}{nV_T}\right)$).

2. **Linear / Triode Region ($V_{GS} \geq V_T$ and $V_{DS} < V_{GS} - V_T$):** An unbroken conducting channel exists from source to drain. The inversion charge density decreases smoothly from source ($Q_n(0) = -C_{ox}(V_{GS} - V_T)$) to drain ($Q_n(L) = -C_{ox}(V_{GS} - V_T - V_{DS})$). Integrating drift current:

$$I_D = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_T)V_{DS} - \frac{1}{2}V_{DS}^2 \right] \tag{11}$$

   For small $V_{DS} \ll 2(V_{GS} - V_T)$, the V_{DS}^2 term is negligible, and the MOSFET acts as a linear **voltage-controlled resistor**:

$$r_{DS} = \left(\frac{\partial I_D}{\partial V_{DS}} \right)^{-1} = \frac{1}{\mu_n C_{ox} \frac{W}{L}(V_{GS} - V_T)} \tag{12}$$

3. **Pinch-Off and Saturation Region ($V_{GS} \geq V_T$ and $V_{DS} \geq V_{DS,\text{sat}} = V_{GS} - V_T$):** When V_{DS} reaches the **saturation voltage** $V_{DS,\text{sat}} = V_{GS} - V_T$, the inversion layer charge at the drain end drops to zero ($Q_n(L) = 0$). This point is called **pinch-off**.

   - Any further increase in V_{DS} cannot increase the voltage drop across the remaining channel; the excess voltage ($V_{DS} - V_{DS,\text{sat}}$) is dropped across the high-field drain depletion region.
   - Electrons reaching the pinch-off point are swept across the depletion space by the high electric field into the drain.
   - Substituting $V_{DS} = V_{GS} - V_T$ yields the square-law saturation equation:

$$I_{D,\text{sat}} = \frac{1}{2}\mu_n C_{ox} \frac{W}{L}(V_{GS} - V_T)^2 \tag{13}$$

3.2 Channel Length Modulation (CLM)

In the saturation region, as V_{DS} increases past $V_{DS,\text{sat}}$, the pinch-off point migrates slightly toward the source. The effective channel length becomes $L_{\text{eff}} = L - \Delta L$. Because $I_D \propto 1/L_{\text{eff}}$, current rises slightly with V_{DS}:

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MOSFET Saturation Current with Channel Length Modulation

$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_T)^2\,(1 + \lambda V_{DS}) \tag{14}$$

where $\lambda = \frac{1}{V_A}$ is the **channel length modulation parameter** (V^{-1}), and V_A is the MOSFET Early voltage ($V_A \propto L$).

4. Small Signal Models of the MOS Transistor

4.1 Low-Frequency Small-Signal Model Parameters

Around a DC operating point $(V_{GS}, V_{DS}, V_{BS}, I_D)$ in the saturation region:

1. **Transconductance (g_m):** Relates drain current variation to gate-to-source voltage changes:

   Three Equivalent Formulas for Transconductance g_m

   $$g_m = \left.\frac{\partial I_D}{\partial V_{GS}}\right|_{V_{DS}} = \mu_n C_{ox}\frac{W}{L}(V_{GS} - V_T) = \sqrt{2\mu_n C_{ox}\frac{W}{L} I_D} = \frac{2I_D}{V_{GS} - V_T} \tag{15}$$

2. **Body Transconductance (g_{mb}):** Accounts for substrate voltage modulation via the body effect:

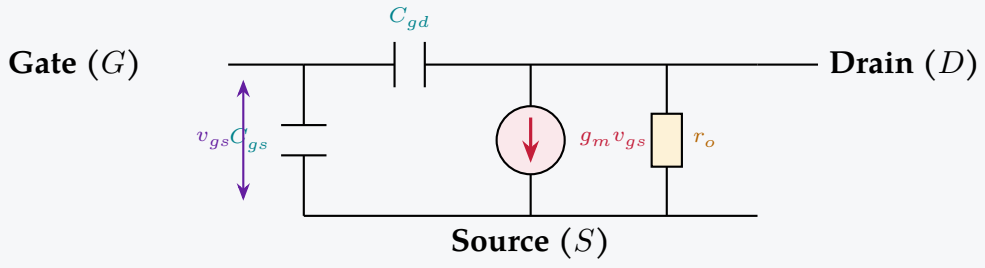
   $$g_{mb} = \left.\frac{\partial I_D}{\partial V_{BS}}\right| = \left(\frac{\partial I_D}{\partial V_T}\right)\left(-\frac{\partial V_T}{\partial V_{SB}}\right) = \frac{\gamma}{2\sqrt{2\phi_F + V_{SB}}} g_m = \eta g_m \tag{16}$$

   where $\eta = \frac{g_{mb}}{g_m} \approx 0.1$–$0.25$.

3. **Output Resistance (r_o):** Arises from channel length modulation:

   $$r_o = \left(\frac{\partial I_D}{\partial V_{DS}}\right)^{-1}\Bigg|_{V_{GS}} = \frac{1}{\lambda I_D} = \frac{V_A}{I_D} \tag{17}$$

Complete Small-Signal Equivalent Circuit of MOSFET (High-Frequency)



4.2 Unity-Gain Cut-Off Frequency (f_T)

The transition frequency f_T is the frequency at which the short-circuit AC current gain of the common-source MOSFET drops to unity ($|i_d/i_g| = 1$):

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Unity-Gain Cut-Off Frequency

$$f_T = \frac{g_m}{2\pi(C_{gs} + C_{gd})} \approx \frac{\mu_n C_{ox} \frac{W}{L}(V_{GS} - V_T)}{2\pi\left(\frac{2}{3}WLC_{ox}\right)} = \frac{3\mu_n(V_{GS} - V_T)}{4\pi L^2} \tag{18}$$

Scaling Law: $f_T \propto 1/L^2$. Reducing physical channel length L yields a quadratic increase in operating speed and device bandwidth.

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Quick Revision — Module 3

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Key Formulas and Metrics at a Glance

Oxide Capacitance per Area	$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$
Flat-Band Voltage (V_{FB})	$V_{FB} = \Phi_{ms} - \frac{Q_{ox}}{C_{ox}}$
Threshold Voltage (V_{T0})	$V_{T0} = V_{FB} + 2\phi_F + \frac{\sqrt{4q\epsilon_s N_A \phi_F}}{C_{ox}}$
Body Effect Threshold Shift	$V_T = V_{T0} + \gamma\left(\sqrt{2\phi_F + V_{SB}} - \sqrt{2\phi_F}\right)$
Linear Region Drain Current	$I_D = \mu_n C_{ox} \frac{W}{L}\left[(V_{GS} - V_T)V_{DS} - \frac{1}{2}V_{DS}^2\right]$
Saturation Drain Current	$I_{D,\text{sat}} = \frac{1}{2}\mu_n C_{ox} \frac{W}{L}(V_{GS} - V_T)^2(1 + \lambda V_{DS})$
Transconductance (g_m)	$g_m = \mu_n C_{ox} \frac{W}{L}(V_{GS} - V_T) = \sqrt{2\mu_n C_{ox} \frac{W}{L} I_D}$
Output Resistance (r_o)	$r_o = \frac{1}{\lambda I_D} = \frac{V_A}{I_D}$
Unity-Gain Frequency (f_T)	$f_T \approx \frac{g_m}{2\pi C_{gs}} \approx \frac{3\mu_n(V_{GS}-V_T)}{4\pi L^2}$
Maximum Depletion Width	$W_{\text{max}} = \sqrt{\frac{4\epsilon_s \phi_F}{qN_A}}$

One-Line Definitions

- **Flat-Band Voltage** (V_{FB})**:** Gate voltage required to negate work-function differences and oxide charges, making bands flat.
- **Strong Inversion:** State where surface potential $\psi_s \geq 2\phi_F$, creating an electron sheet equal to bulk hole doping.
- **Pinch-Off:** Phenomenon where inversion channel charge at the drain vanishes when $V_{DS} \geq V_{GS} - V_T$.
- **Body Effect:** Increase in threshold voltage caused by applying a reverse bias between source and body substrate.
- **Channel Length Modulation:** Shortening of the effective channel length under high V_{DS}, producing finite output resistance.
- **Transconductance** (g_m)**:** Rate of change of drain current with respect to gate-source voltage at constant V_{DS}.
- **Transition Frequency** (f_T)**:** Frequency at which the small-signal short-circuit current gain drops to unity ($0\,\text{dB}$).

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Frequently Asked / Viva Questions

Q1. Why does the high-frequency C-V curve of an MOS capacitor remain at C_{min} in inversion while the low-frequency curve recovers to C_{ox}?

In an isolated MOS capacitor, minority electrons can only be supplied via thermal generation-recombination across the bandgap, which is slow ($\tau \sim 0.01$–$0.1\,\text{s}$). At high frequencies ($f \geq 100\,\text{kHz}$), minority electrons cannot follow the AC signal, so the depletion layer capacitance $C_{\text{dep,max}}$ remains in series with C_{ox}, locking total capacitance at C_{min}. At low frequencies ($f \leq 100\,\text{Hz}$), minority carriers respond, shielding the silicon bulk and restoring $C = C_{ox}$.

Q2. Why does a MOSFET invert instantaneously while a standalone MOS capacitor takes significant time?

A MOSFET has heavily doped n^+ Source and Drain regions immediately adjacent to the channel, providing an abundant reservoir of electrons that can flow into the channel within picoseconds, whereas an MOS capacitor must wait for slow thermal EHP generation.

Q3. What is the physical condition for the onset of strong inversion?

The surface potential must equal twice the bulk Fermi potential: $\psi_s = 2\phi_F = 2V_T \ln(N_A/n_i)$. At this potential, the surface electron concentration $n_s = n_i e^{(\psi_s - \phi_F)/V_T}$ equals the substrate acceptor concentration N_A.

Q4. Explain the physical mechanism of the Body Effect in a MOSFET.

Applying a reverse substrate bias ($V_{SB} > 0$) widens the depletion layer beneath the gate, exposing additional negative acceptor ions ($-qN_A$). The gate voltage must support this additional depletion charge before creating the inversion layer, thereby increasing V_T.

Q5. What is Channel Length Modulation (CLM) and what is its effect on amplifier gain?

In saturation, high V_{DS} shifts the pinch-off point toward the source, decreasing effective channel length $L_{\text{eff}} = L - \Delta L$. This causes drain current to increase with V_{DS}, giving the MOSFET a finite output resistance $r_o = 1/(\lambda I_D)$ and limiting maximum intrinsic voltage gain $A_{v0} = g_m r_o$.

Q6. Compare enhancement-mode and depletion-mode MOSFETs.

Enhancement-mode MOSFETs have no physical channel at $V_{GS} = 0$ (normally OFF) and require $V_{GS} > V_T > 0$ to conduct. Depletion-mode MOSFETs have an implanted physical channel at $V_{GS} = 0$ (normally ON) and require a negative gate voltage $V_{GS} < V_T < 0$ to deplete the channel and turn off.

Q7. Why is an n-channel MOSFET faster and smaller than a p-channel MOSFET for the same drive current?

Electron mobility in Silicon ($\mu_n \approx 1350\,\text{cm}^2/\text{V}\cdot\text{s}$) is nearly three times higher than hole mobility ($\mu_p \approx 480\,\text{cm}^2/\text{V}\cdot\text{s}$). To achieve equal current ($I_D \propto \mu W/L$), a PMOS device must be designed with roughly 2.5–3 times greater width W, which increases parasitic capacitance and area.

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Frequently Asked / Viva Questions (contd.)

Q8. How does the unity-gain frequency f_T depend on MOSFET gate length?

$f_T \approx \frac{3\mu_n(V_{GS}-V_T)}{4\pi L^2} \propto \frac{1}{L^2}$. It scales inversely with the square of the channel length, which has been the primary technological driving force behind semiconductor feature scaling.

Q9. What is deep depletion in an MOS capacitor?

Deep depletion occurs when a rapid positive voltage ramp is applied to the gate so fast that no inversion electrons can generate. The depletion layer expands past W_{max}, and capacitance continues dropping below C_{min}. This state is utilized in CCD image sensors.

Q10. What is the transconductance parameter k_n' and device transconductance parameter k_n?

Process transconductance parameter $k_n' = \mu_n C_{ox}$ (A/V^2), which depends purely on fabrication technology. Device transconductance parameter $k_n = k_n'(W/L) = \mu_n C_{ox}(W/L)$, which depends on circuit designer sizing.

Comparison of BJT and MOSFET Device Characteristics

Parameter	Bipolar Junction Transistor (BJT)	MOSFET
Operating Principle	Current-controlled device ($I_C = \beta I_B$)	Voltage-controlled device ($I_D \propto V_{GS}^2$)
Carrier Type	Bipolar (both electrons and holes participate)	Unipolar (majority carriers only in channel)
Input Impedance	Low to moderate (1–10 kΩ in active mode)	Extremely high ($> 10^{12}$ Ω, isolated gate oxide)
Input Current	Continuous DC base current $I_B = I_C/\beta$	Zero static gate current ($I_G \approx 0$)
Temperature Coefficient	Negative (NTC): prone to thermal runaway	Positive (PTC): thermally self-limiting
Noise Figure	Low $1/f$ flicker noise; higher shot noise	Higher flicker noise ($1/f$ interface states)
Fabrication Density	Moderate; requires isolation wells	Extremely high; self-aligned planar process
Primary Applications	High-speed RF, precision analog, linear power	Digital VLSI, microprocessors, CMOS logic, power

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